

CPP NOTES – DAY 27

Classes and Objects

- Class name must start with an uppercase letter(not mand). If class has two words, then first should be uppercase. Eg: class Study, class StudyTonight etc.
- The access of data members and methods depends upon the access specifiers.
- The members are private for default in class and that is the difference between structure and classes in cpp.
- Objects are initialized using special class functions called **constructors**.
- Object in cpp is identified as a class not like how a variable is identified(as single data type)

Access Control

- Defines the control rules in cpp
- Private, protected and public
- Private – default(only available inside the class)
- Protected – behaves like private but child class can inherit.
- Public – can access from outside.
- If I declare a public variable inside class and created a pointer for the instance of the class, the value can be updated and modified. So to avoid that, we can make it private to not get accessed outside.
- To access private variables outside, we need interfaces to connect with the variables.
- Message passing – communication between objects
- Run method is used to configure the class to certain values.
- Init method is used to initialize values to variables inside the class.

Functions

1. Simple function - regular member function of a class. It can access both private and public members of the class.
2. Static function – when a class is created, the class declaration will be in the text segment, but the static member function will occupy a space in the stack.
3. Const function – cannot modify, only read.
4. Inline function – expression evaluated at the compile time.
5. Friend function – can access methods inside the class from outside. Have to declare this function inside the class with friend keyword.

Constructors

Are of three types:

1. Default constructor
2. Parameterized constructor
3. Copy constructor

FYI

- *Why constructors cannot be declared private? Ans: If we declare constructor in private, whenever we are initializing an object of the class, since the constructor also should be initiated, it cant access the constructor due to private declaration. So to avoid that conflict , it is declared in public.*